

Beating YOLO Decisively

The model already leads on every metric. This is the plan to turn that lead into a claim that survives scrutiny — and to close the one gap that is genuinely large.

Model V6.2, 6.42 M parameters, trained from scratch Status 120/120 epochs complete

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01 Where it stands, measured

MASK METRIC	V6.2 VAL	V6.2 TEST	YOLO11L-SEG
mAP50-95	0.8301	0.7499	0.6932
mAP50	0.9747	0.9749	0.9693
mAP75	0.9434	0.8498	not reported
precision	0.9310	0.9639	0.9543
recall	0.9751	0.9812	0.9120
parameters		6,417,683	~27,600,000

Complete splits, final checkpoint, after the decoder fix in step 1.
YOLO11l-seg is its own best epoch on its own validation split.

The lead is real but not yet defensible. That YOLO baseline stopped at **epoch 23 of 300** — 8% of its schedule, still at roughly 92% of its peak learning rate — because 300 epochs at 74.9 min/epoch is 15.6 days on this laptop.

A properly trained YOLO will score meaningfully above 0.693. Steps 6 and 7 exist to find out by how much, instead of quietly benefiting from the comparison.

02 Where the remaining loss actually is

The test split answers this precisely, and the answer is not what the headline number suggests. Detection is finished; **outlining is not**.

test AP50 – can it find the object?



nothing is being missed, in any class

test AP50-95 – can it outline the object?



the whole 0.225 gap is mask geometry, concentrated in the smallest objects

Bars are drawn to a common 0–1 scale. The distance between the two is what every step below is aimed at.

CLASS	SHARE	AP50	AP50-95	GAP
Rectangle	45%	0.9892	0.8159	0.173
Rectangle_concave	0.3%	0.9256	0.8105	0.115
circle_full	45%	0.9901	0.7959	0.194
circle (annulus, 16 px)	10%	0.9821	0.5630	0.419

Per class on test. The AP50-95 ordering is the object-size ordering, inverted: median diameters run 108 px for Rectangle_concave down to 16 px for circle and circle_full.

circle is an annulus whose median diameter falls from 33.8 px in training to **16.1 px** in test — a ring three or four pixels thick. It is found 98% of the time and outlined tightly almost never. That single class is the largest recoverable loss in the model.

03 The plan

01 Stop minting instances from offset residue DONE

no retraining · measured, not estimated

The decoder turned every foreground pixel whose predicted offset landed near no centre into a new instance. Measured over 150 validation images by re-running the official matcher with and without that path: **75 false positives against 2 true positives**. The residue is object rims and specular highlights, not missed objects.

Result on the complete splits, no retraining: test mAP50-95 0.7463 → **0.7499**, precision 0.9595 → **0.9639**, 155 fewer false positives and two *more* true positives.

Rectangle_concave precision on test 0.4414 → **0.7000**.

02 Dihedral test-time augmentation BUILT, NOT YET MEASURED

no retraining · 8× inference · targets mask geometry

Every class here is symmetric under all eight flips and quarter turns, so their predictions can be averaged. Averaging eight views sharpens exactly what is weak: the sub-pixel placement of a boundary on a 16-pixel ring.

The reason this is a module and not a loop: **the offset head is a vector field**. Un-rotating the image grid is only half the inverse; every (dx, dy) pair has to be rotated too, or the averaged field points eight ways and the decoder groups nothing. Its self-check already caught one real aliasing bug that made exactly one of the eight transforms silently wrong.

03

Fit the five decoder thresholds NEXT

no retraining · hours · split-validated

The thresholds in use are the ones Model V5 was tuned with, and they are shared across all four classes. Random search over 200 draws, scored on cached validation predictions, **fitted on one half of validation and verified on the other** — otherwise it fits noise.

One caveat has to travel with the result: 44 images are byte-identical between validation and test, so anything tuned on validation contaminates roughly 10% of the test set. That must be stated wherever the test number is quoted.

04

Predict masks at full resolution RETRAIN

~3 days · the structural fix for the 0.225 gap

Both decoders currently stop at 256×256 and the logit maps are bilinearly upsampled to 512. That was a deliberate speed trade — 70.6% of the model's 42.75 GMAC sit at that resolution — and its cost is written into the note that introduced it: *slightly softer boundaries*.

A 16-pixel annulus is 8 pixels at head resolution and its ring is under two. That is where AP50-95 0.5630 comes from. Restoring full resolution for the semantic and boundary heads is the direct fix; it will not fit in 8 GB at batch 2 without also cutting the centre/offset heads, which step 5 does anyway.

05

Replace centre+offset with distance-transform watershed RETRAIN

~3 days · frees 22.6% of the compute

Centre+offset grouping comes from Panoptic-DeepLab, built for crowded, occluded street scenes. These are well-separated pads. It is the most data-hungry part of the model and the two head stacks alone are **22.6% of all MACs**, both at 256×256.

Swapping the two-component vector field for one scalar inner-distance field is easier to learn, removes the residue problem in step 1 by construction, and buys back the compute that step 4 needs.

06

Train YOLO properly

2-3 DAYS

the honest baseline

Resume from `last.pt` at epoch 24 with its optimiser state and take it to at least epoch 80, so both models are compared at a similar fraction of their schedules rather than 100% against 8%.

07

One protocol, one test set, error bars

1 DAY

what turns the number into a claim

Score both models on the same 432 test images with the same mAP implementation, and bootstrap that set 1,000 times for 95% intervals. With 432 images the difference between 0.75 and 0.77 may sit inside the sampling noise; overlapping intervals would mean nothing has been beaten.

Also fix the labels: 19.4% of the test split contains polygons that overwrite one another, which caps the IoU any model can reach.

04 Why the architecture should win, and where it should not

Three differences are structural rather than a matter of training length, and all three act at exactly the IoU thresholds where the gap sits.

Prototype masks are low-rank. YOLO-seg reconstructs each instance as a weighted combination of a small basis of prototypes at reduced resolution. On a 16-pixel disc the rim *is* the object, and the rim is what a shared basis cannot spend capacity on. V6.2 predicts a class per pixel at full resolution; no mask is constrained to resemble a blend of others.

A mask cropped to a box inherits the box's error. YOLO-seg multiplies its mask by the detected bounding box. Slightly loose admits background; slightly tight shaves the object. At IoU 0.5 that rarely changes the verdict; at 0.85 it decides it. V6.2 produces no boxes at all.

Axis-aligned boxes describe a rotated pad badly. These pads sit at arbitrary angles — the training cycle alone sweeps 45 rotations. A box around a diagonal elongated pad is mostly background, which weakens both the mask crop and the non-maximum suppression deciding which detections survive. Dense per-pixel grouping has no orientation to lose.

Against that, two honest disadvantages. YOLO arrives pretrained on COCO and so carries a transferable prior to a new board type, where V6.2 — 4,680 images from random initialisation — carries none. And YOLO is far cheaper at inference: V6.2 currently decodes at 1.5 images/second, most of it CPU-side instance grouping.

05 What success looks like

AFTER	TEST MAP50-95	COST	RETRAIN
today	0.7499	—	—
+ TTA (step 2)	0.76 – 0.78	hours	no
+ fitted thresholds (step 3)	0.78 – 0.80	hours	no
+ full-resolution masks (4, 5)	0.82 – 0.86	~3 days	yes
+ repaired labels (7)	raises the ceiling itself	days	yes

Targets, not predictions. Steps 1–3 need no retraining; the figures for 4 and 5 are the project's own prior estimates.

The target is not a bigger number. It is a number on an untouched test split, under one protocol, against a fully-trained YOLO, with confidence intervals that do not overlap.

Everything above either moves that number or makes it mean something.

Model V6.2 · 6,417,683 parameters · trained from random initialisation on 4,680 images; no pretrained weights, nothing downloaded.
All V6.2 figures are measured on the complete 1,048-image validation and 432-image test splits.
YOLO11l-seg figures are read from its own results.csv at its best epoch (22 of a planned 300).